

Exercises on Java synchronization
NOT ASSESSED

Instructions: This exercise sheet is not assessed - it is to help you to understand the material, and to challenge what you think you know. Solutions for the following questions will be provided.

Collaboration: You are permitted to work with your classmates. However, you are likely to get more out of this if you attempt the answers first, before discussing them.

Consider the following class.

```
public class Example {
    private int x = 2;
    private int y = 3;

    public void a() {
        y = 2 * x;
    }

    public synchronized void b() {
        x = x + y;
    }

    public synchronized void c() {
        y = 2 * x;
    }

    public synchronized void d() {
        System.out.println("The values of x and y are " + x + " and " + y);
    }
}
```

1. Suppose that two threads T1 and T2 run concurrently, and T1 tries to execute method a() while T2 tries to execute method b() on the same instance of the Example class.
 - (a) Can a() and b() be executed concurrently by T1 and T2? Motivate your answer.
 - (b) What are the possible values of x and y after T1 and T2 terminate?
2. Consider now two threads T3 and T4 that run concurrently, and such that T3 tries to execute method b() while T4 tries to execute method c() on the same instance of the Example class.
 - (a) Can b() and c() be executed concurrently by T3 and T4? Motivate your answer.

- (b) What are the possible values of `x` and `y` after `T3` and `T4` complete the execution of `b()` and `c()` respectively?
3. Can `b()` and `c()` be executed concurrently on different instances of the `Example` class? That is, can a thread `T5` execute `b()` on one instance of the `Example` class, while another thread `T6` executes `c()` on another `Example` instance? Justify your answer explicitly referring to the semantics of the `synchronized` keyword.
4. Is the `Example` class thread-safe? Motivate your answer.